

Git & Github

- Git is a Version Control System. → is a tool that helps to track changes in code
- It is :
- popular
 - free & Open Source
 - fast & Scalable

- (1) Track the history
 - (2) Collaborate
- } ⇒ Git

- Github → website that allows developers to store and manage their code using Git

folder (Repository)

<https://github.com>

- commit → change

Setting up Git

VS code

Windows (Git Bash)

git --version

Configuring Git

git config --global user.name "Myname"

git config --global user.email "someone@email.com"

git config --list

Clone & Status

Clone - Cloning a repository on our local machine

git clone <- somelink ->

status - displays the status of the code

git status

cd → change directory

cd .. ⇒ back from directory

untracked → new files that git doesn't yet track

modified → changed

staged → file is ready to be committed

unmodified → unchanged.

mkdir LocalRepo → folder बनाइ नाम "LocalRepo"

Add & Commit

add - adds new or changed files in your working directory to the Git staging area

git add <file name>

commit - it is the record of change

git commit -m "some message"

Note : all files add ~~not~~ → git add .

Push Command

push - upload local repo content to remote repo

git push origin main

Init Command

init - used to create a new git repo

git init

git remote add origin <link>

git remote -v (to verify remote)

git branch (to check branch)

git branch -M main (to rename branch)

git push origin main

Workflow

local Git

GitHub repo

↓
clone

↓
change

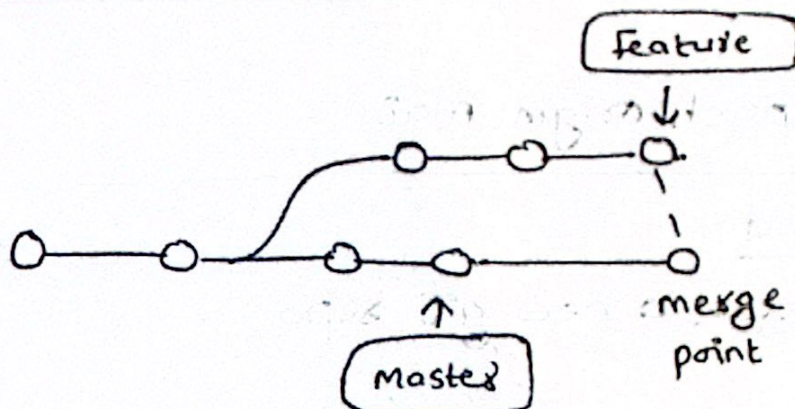
↓
add

↓
commit

↓
push



Git Branches



Branch Commands

git branch (to check branch)

git branch -M main (to rename branch)

git checkout <branch name> (to navigate)

git checkout -b <new branch name> (to create new branch)

git branch -d <branch name> (to delete branch)

Merging Code

Way 1 :

git diff <branch name> (to compare commits, branches, files & more)

git merge <branch name> (to merge 2 branches)

Way 2 :

Create a PR

Pull Command

`git pull origin main`

- used to fetch and download content from a remote repo and immediately update the local repo to match that content.

Resolving Merge Conflicts

An event that takes place when Git is unable to automatically resolve differences in code between two commits,

- remove <<<<<<
- =====
- feature1 HT
- git add.
- git commit -m "add both features"
- git checkout main
- git merge feature1
- git push origin main

Undoing Changes

Case 1: staged changes

`git reset <-file name->`

`git reset`

Case 2 : committed changes (for one commit)

`git reset HEAD~1`

Case 3 : committed changes (for many commits)

`git reset <-commit hash->`

`git reset --hard <-commit hash->`

Note : `git log` મા જાને જો hash code
(history આડક)

Fork

A fork is a new repository that shares code and visibility settings with the original "upstream" repository.

Fork is a rough copy.